

EDUCATION

Georgia Institute of Technology Postdoctoral Fellow: Mathematics	Atlanta, USA 2026–TBD
– Research area: <i>Stochastic mixed-variable optimization and machine learning</i> – Supervision : Pr. Katya Scheinberg	
Polytechnique Montréal Ph.D.: Applied mathematics	Montréal, QC 2022–2026
– Thesis: <i>Contributions to constrained mixed-variable and hierarchical blackbox optimization</i> – GPA : 4/4 (17 credits) – Supervision: Pr. Charles Audet, Pr. Sébastien Le Digabel and Pr. Youssef Diouane	
Polytechnique Montréal M.Sc.A : Applied mathematics	Montréal, QC 2020–2022
– Master's thesis: <i>A general mathematical framework for constrained blackbox optimization problems with meta and categorical variables</i> – GPA: 4/4 (18 credits) – Supervision: Pr. Charles Audet and Pr. Sébastien Le Digabel	
Polytechnique Montréal B.Ing: Engineering physics	Montréal, QC 2016–2020
– Specialization: Mathematics – GPA: 3.84/4 (126 credits) – Experience : two research internships funded by NSERC-FRQNT	

SCHOLARSHIPS AND AWARDS

Postdoctoral research	
• FRQNT Postdoctoral Fellowship: \$110,000 over two years	2026–2028
PhD	
• Nomination for the Best Thesis in Applied Mathematics at Polytechnique Montréal • FRQNT scholarship: \$100,000 over four years (partially held concurrently with NSERC) • NSERC scholarship: \$63,000 over three years • Trottier Energy Institute scholarship: \$51,000 over three years • GERAD conference grant: \$1,750 (ICIAM 2023 conference in Japan)	2026 2023–2027 2023–2026 2022–2025 2023
Master's	
• Nomination for the Best Master's Thesis in Applied Mathematics at Polytechnique Montréal • IVADO Excellence Scholarship: \$40,000 over two years • Hydro-Québec Excellence Scholarship: \$10,000 over one year	2022 2020–2022 2020–2021

Bachelor's

- NSERC Undergraduate Student Research internship award: \$4,500 over one term 2019
- FRQNT Undergraduate Research internship scholarship: \$2,000 over one term 2019
- NSERC Undergraduate Student Research internship award: \$4,500 over one term 2018
- FRQNT Undergraduate Research internship scholarship: \$2,000 over one term 2018

PUBLICATIONS

Authors are listed in alphabetical order in mathematical optimization journals. I am the first author on all papers except [2], for which I am the second author.

- [1] C. Audet, Y. Diouane, E. Hallé-Hannan, S. Le Digabel and C. Tribes (2026). “Surrogate-based categorical neighborhoods for mixed-variable blackbox optimization”. *In revision at Journal of Global Optimization (JOGO)*.
- [2] P. Saves, E. Hallé-Hannan, J. Bussemaker, Y. Diouane and N. Bartoli (2026). “Modeling hierarchical spaces: a review and unified framework for surrogate-based architecture design”. *Structural and Multidisciplinary Optimization* 69(3), 65. DOI: 10.1007/s00158-026-04249-2
- [3] C. Audet, Y. Diouane, E. Hallé-Hannan, S. Le Digabel and C. Tribes (2025). “CatMADS: Mesh adaptive direct-search for constrained blackbox optimization with categorical variables”. *To appear in SIAM Journal of Optimization (SIOPT)*. PRE-PRINT: arXiv.
- [4] E. Hallé-Hannan, C. Audet, Y. Diouane, S. Le Digabel and C. Tribes (2025). “Cat-Suite: A collection of optimization problems with categorical and quantitative variables for benchmarking”. Technical report, *Les cahiers du GERAD*. Manuscript: G-2025-39.
- [5] E. Hallé-Hannan, C. Audet, Y. Diouane, S. Le Digabel and P. Saves (2025). “A distance for mixed-variable and hierarchical domains with meta variables”. *Neurocomputing* 653, 131208. DOI: 10.1016/j.neucom.2025.131208
- [6] C. Audet, E. Hallé-Hannan et S. Le Digabel (2023). “A General Mathematical Framework for Constrained Mixed-variable Blackbox Optimization Problems with Meta and Categorical Variables”. *Operations Research Forum* 4.12. DOI: 10.1007/s43069-022-00180-6

SEMINARS

- [1] *CatMADS: categorical variables with the MADS algorithm – Invited seminar (1h) à l’Institut de Recherche en Informatique de Toulouse (UT Capitole)*. Seminar in Toulouse (December 2025).

CONFERENCE PRESENTATIONS

- [1] *Handling Categorical Variables in Mixed-Variable Trust-Region Optimization – 26th International Symposium on Mathematical Programming (ISMP)*. International conference in Amsterdam (to be held in July 2027).
- [2] *NomadBBO: A Python library for constrained mixed-variable blackbox optimization – SIAM Conference on Computational Science and Engineering (CSE27)*. International conference in Pittsburgh (to be held in February 2027).

- [3] *Surrogate-based categorical neighborhoods in CatMADS – Optimization 2026* organized by the Portuguese Operations Research Society (APDIO). International conference in Portugal (July 2026).
- [4] *Surrogate-based categorical neighborhoods in CatMADS – Third Derivative-Free Optimization Symposium*. International conference in Montreal (June 2026).
- [5] *Surrogate-based categorical neighborhoods in CatMADS – Optimization Days*. International conference in Montreal (May 2026).
- [6] *CatMADS: categorical variables with the MADS algorithm and surrogate-based neighborhoods – INFORMS Optimization Society Conference*. International conference in Atlanta (March 2026).
- [7] *CatMADS: categorical variables with the MADS algorithm – Canadian Mathematical Society Summer Meeting*. International conference in Quebec City (June 2025).
- [8] *CatMADS: categorical variables with the MADS algorithm – Optimization Days (JOPT)* organized by GERAD and CIRRELT. International conference in Montreal (May 2025).
- [9] *A graph-structured distance for heterogeneous datasets with meta variables – 25th International Symposium on Mathematical Programming (ISMP)*. International conference in Montreal (July 2024).
- [10] *A graph-structured distance for heterogeneous datasets with meta variables – Optimization Days*. International conference in Montreal (May 2024).
- [11] *A General Blackbox Optimization Framework for Hyperparameter Optimization in Deep Learning – 10th International Congress on Industrial and Applied Mathematics (ICIAM 2023 Tokyo)*. International conference in Tokyo (August 2023).
- [12] *A General Blackbox Optimization Framework for Hyperparameter Optimization in Deep Learning – Optimization Days (JOPT)* organized by GERAD and CIRRELT. International conference in Montreal (May 2023).
- [13] *A General Blackbox Optimization Framework for Hyperparameter Optimization in Deep Learning – SIAM Conference on Computational Science and Engineering (CSE23)*. International conference in Amsterdam (March 2023).
- [14] *Notation framework for hyperparameter tuning in deep learning – Optimization Days*. International conference in Montreal (May 2022).
- [15] *Cadre mathématique pour l’optimisation des hyperparamètres en I.A. – Octobre numérique* organized by IVADO. Provincial conference (October 2021).

SUPERVISION

1. Noam Charette (*2nd year, Electrical Engineering*) for an NSERC–FRQNT undergraduate research internship at Polytechnique Montréal and GERAD. *Development of hyperparameter optimization problems for mixed and hierarchical optimization* – May 2025 to August 2025.
2. Elie Boulanger (*4th year, Computer Engineering*) for an undergraduate research internship at Polytechnique Montréal and GERAD. *Design of realistic blackboxes for hyperparameter optimization with categorical variables* – May 2025 to August 2025.
3. Amaury Diopus’kin (*Master 2, France*) for a research internship at GERAD. *Implementation of a code architecture to address hyperparameter optimization problems in deep learning* – April 2022 to September 2022.

TEACHING EXPERIENCE

Department of Mathematics and Industrial Engineering, Polytechnique Montréal

Montreal, QC

Lecturer

Teach theory sessions and prepare course materials.

- MTH1115 – Differential Equations (W2026)

Lab Instructor

Teaching laboratory sessions: theoretical exercises and computer lab sessions.

- MTH1115 – Differential Equations (F2025, W2026)
- MTH2302B – Probability and Statistics (S2022)
- MTH2402 – Operations Research (W2022)
- MTH2312 – Advanced Statistical Methods (W2022)

Teaching Assistant

- MTH8418 – Derivative-Free Optimization (W2022, W2023, W2024, W2025)
 - Develop and supervise a course project on challenging blackbox optimization
 - Teach two to three lectures related to the project
 - Grade the project across two deliverables

ACADEMIC SERVICE

Reviewer

- Mathematical Programming
- Journal of Optimization Theory and Applications
- Journal of Global Optimization
- Computational Optimization and Applications
- Optimization Methods and Software

Session Chair

- 25th International Symposium on Mathematical Programming (ISMP): 2024
- Optimization Days (JOPT): 2022 –2026

Volunteering

- 25th International Symposium on Mathematical Programming (ISMP): participant reception, room support, and logistical support, 2024